CertSwap

Intellectual Property Declaration & Proof of Concept

Declaration

This document constitutes a formal declaration of intellectual property ownership and proof of concept for the CertSwap platform. The concepts, architecture, and innovations described herein were conceived and developed by the author identified below. This document is submitted for blockchain timestamping to establish an immutable record of prior art and ownership.

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Public Frontend Showcase: github.com/certswap/certswap-app

Core Concept

CertSwap is a blockchain-based digital certificate, loyalty, and anti-counterfeit platform built on Polygon. The platform introduces a novel architecture combining three operational modules into one unified product, allowing brands to issue tokenized obligations to customers without requiring crypto knowledge, while simultaneously enabling native Web3 functionality for technically sophisticated users.

The Fundamental Innovation:

A token in this system represents a brand's obligation — not a speculative asset. The blockchain functions as a notary and registry, not as an investment vehicle. This distinction allows CertSwap to operate in regulatory grey zones where crypto speculation is restricted, while providing the security, transparency, and immutability of blockchain technology.

Three-Module Architecture

Module 1 — Web3 / DeFi (M1)

Non-custodial mode. Every transaction is signed directly by the user's MetaMask wallet on Polygon. Features: P2P token exchange between brands via on-chain orderbook with escrow, gift certificates as on-chain transactions, payment QR tokens confirmed by authorized merchants, anti-counterfeit via unique QR hashes recorded in smart contract.

Module 2 — Backend / B2B Loyalty (M2)

Blockchain-invisible mode. The system wallet signs transactions on behalf of users. User identification via email or UUID — no MetaMask required. Features: same as M1 but blockchain is hidden from end user, suitable for FMCG, HoReCa, retail loyalty programs, offline-resistant QR payment tokens, duplicate scan detection for anti-counterfeit.

Module 3 — Bridge (M3)

Novel innovation: both M1 and M2 operate simultaneously. At each signing transaction, the user chooses how to store the record: in their personal wallet (Web3) or in the system as a notary (Backend). Value transfer between worlds occurs via gift certificate mechanism. eIDAS-compatible: bridge transactions constitute legally significant acts under EU Regulation 910/2014 on electronic trust services.

Non-Obvious System Capabilities

• Anti-counterfeit via bonus mechanic:

The user unknowingly verifies product authenticity by scanning QR to receive a bonus. Duplicate scans are detected and logged with IP, geolocation, and device fingerprint. The brand receives a counterfeit signal without user awareness of the security function.

• Offline-resistant QR tokens:

Payment tokens can be presented without internet connection. Confirmation occurs when connectivity is restored. The obligation exists physically on the QR regardless of network availability.

• Privacy threshold design:

User identity is tied to device/email, not passport. Sufficient for legal compliance, insufficient for criminal anonymity. This design choice is intentional and constitutes a novel approach to regulatory compliance.

• Legal decentralized value transfer:

Enables transfer of value between parties without a bank, without crypto knowledge, and without speculative assets. Tokens represent brand obligations, not financial instruments.

• Bridge as eIDAS notarial act:

When a user transfers value between M1 and M2, the bridge transaction is recorded as a notarial act in the blockchain, meeting eIDAS requirements for electronic trust services.

Technical Implementation

• Smart Contract: Solidity 0.8.20, deployed on Polygon Amoy Testnet (proof of concept; contract address subject to change as the platform evolves toward mainnet — this declaration covers the underlying concepts and architecture independent of any specific contract deployment)

• Frontend: Vanilla HTML/CSS/JS, ethers.js, deployed on Vercel

• Backend: Node.js/Express, PostgreSQL

• Live Demo: https://certswap.vercel.app

• Public Frontend Showcase: github.com/certswap/certswap-app

Declared Future Innovations (Roadmap)

The following concepts are declared as part of this IP, pending implementation:

• Account Abstraction (ERC-4337) — email wallet without MetaMask, tokens truly owned by user

• Module 4 — Blockchain Bill of Exchange: B2B tokenized obligations between companies with eIDAS legal force

• Exchange rate oracle for non-1:1 P2P token swaps

• DAO governance of platform parameters

• Secondary token market / DEX for brand certificates

• White-label deployment for enterprise partners

• Multi-chain support (Base, Arbitrum, Ethereum mainnet)

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